



**VISHWAKARMA
UNIVERSITY**
Maximising Human Potential

Chalo – Live Bus Tracking App

Activity based

Project Report on

Software Engineering

Project Module - I

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Project Name : Sentiment Analysis for Customer Reviews

Introduction:

In Phase 2 of the project, our primary focus was on preparing our predictive model for practical implementation. We meticulously evaluated various machine learning algorithms using a pre-processed CSV file containing customer reviews data. Our goal was to select the most appropriate model that could accurately predict customer sentiments. Additionally, we devoted efforts to setting up the necessary systems to ensure the seamless integration and real-time functionality of the model. Concurrently, we dedicated time to designing the user interface and refining the technical aspects to ensure a smooth user experience.

Model:

During the evaluation phase, we assessed the performance of three distinct machine learning algorithms

1)Random Forest: This algorithm demonstrated exceptional performance, achieving an accuracy, precision, recall, and F1-score of 1.0. Its consistent and flawless predictions across all metrics reaffirmed its suitability for our task.

2)Support Vector Machine (SVM): While not perfect, SVM exhibited strong performance with an accuracy of 0.963. It delivered respectable scores for precision, recall, and F1-score, indicating its reliability in predicting customer sentiments.

3)K-Nearest Neighbors (KNN): KNN also performed remarkably well, with an accuracy close to 1.0 and high scores for precision, recall, and F1-score. Its effectiveness in making accurate predictions based on nearest neighbors was evident.

Upon deploying these models, we obtained the following results:

1)Random Forest: The model achieved an accuracy of 0.8998, showcasing robust performance across all sentiment categories. Notably, it excelled in predicting positive sentiments.

Score:

Accuracy: 0.8998317524566579

Classification Report:

	precision	recall	f1-score	support
negative	0.85	0.69	0.76	5557
neutral	0.20	0.00	0.00	2033
positive	0.91	0.99	0.95	33421
accuracy			0.90	41011
macro avg	0.65	0.56	0.57	41011
weighted avg	0.86	0.90	0.87	41011

Test:

```
new_reviews = ["This product is amazing!"]
new_reviews_tfidf = tfidf_vectorizer.transform(new_reviews)
predicted_sentiments = model.predict(new_reviews_tfidf)
print("Predicted Sentiments:", predicted_sentiments)
```

Predicted Sentiments: ['positive']

2)SVM: With an accuracy of 0.8999, the SVM model exhibited consistent performance across sentiment categories, albeit struggling slightly in predicting neutral sentiments.

Accuracy: 0.8999292872643925

Classification Report:

	precision	recall	f1-score	support
negative	0.85	0.69	0.76	5557
neutral	0.00	0.00	0.00	2033
positive	0.91	0.99	0.95	33421
accuracy			0.90	41011
macro avg	0.58	0.56	0.57	41011
weighted avg	0.85	0.90	0.87	41011

3)KNN: The KNN model achieved an accuracy of 0.8883, demonstrating strong predictive capabilities, particularly in discerning positive sentiments.

Accuracy: 0.8883470288459194

Classification Report:

	precision	recall	f1-score	support
negative	0.85	0.68	0.76	5557
neutral	0.22	0.14	0.17	2033
positive	0.92	0.97	0.94	33421
accuracy			0.89	41011
macro avg	0.66	0.60	0.62	41011
weighted avg	0.87	0.89	0.88	41011

Test:

```
new_reviews = ["This product is amazing!"]
new_reviews_tfidf = tfidf_vectorizer.transform(new_reviews)
predicted_sentiments = model.predict(new_reviews_tfidf)
print("Predicted Sentiments:", predicted_sentiments)
```

Predicted Sentiments: ['positive']

Overall, our models yielded promising results, underscoring their efficacy in predicting customer sentiments based on reviews. These findings laid a solid foundation for further optimization and deployment in real-world applications.

Pseudo Code:

1. Random Forest Classifier:

- Import necessary libraries for data manipulation, model selection, and evaluation.
- Load the customer reviews dataset from a CSV file.
- Preprocess the dataset, handling any missing values.
- Define the features (reviews) and target variable (sentiment).

- Perform text preprocessing, such as TF-IDF vectorization.
- Initialize a Random Forest Classifier.
- Split the dataset into training and testing sets.
- Train the Random Forest model on the training data.
- Make predictions on the test data.
- Evaluate the model's performance using `accuracy_score` and `classification_report`.
- Output the accuracy and classification report.

2. Support Vector Machine (SVM) Classifier:

- Import necessary libraries for data manipulation, model selection, and evaluation.
- Load the customer reviews dataset from a CSV file.
- Preprocess the dataset, handling any missing values.
- Define the features (reviews) and target variable (sentiment).
- Perform text preprocessing, such as TF-IDF vectorization.
- Initialize an SVM Classifier.
- Split the dataset into training and testing sets.
- Train the SVM model on the training data.
- Make predictions on the test data.
- Evaluate the model's performance using `accuracy_score` and `classification_report`.
- Output the accuracy and classification report.

3. K-Nearest Neighbors (KNN) Classifier:

- Import necessary libraries for data manipulation, model selection, and evaluation.
- Load the customer reviews dataset from a CSV file.
- Preprocess the dataset, handling any missing values.
- Define the features (reviews) and target variable (sentiment).
- Perform text preprocessing, such as TF-IDF vectorization.
- Initialize a KNN Classifier.

- Split the dataset into training and testing sets.
- Train the KNN model on the training data.
- Make predictions on the test data.
- Evaluate the model's performance using `accuracy_score` and `classification_report`.
- Output the accuracy and classification report.

These pseudocode snippets outline the step-by-step process for training and evaluating each machine learning model (Random Forest, SVM, and KNN) for sentiment analysis of customer reviews. Each snippet covers data preprocessing, model initialization, training, prediction, evaluation, and output of results.

Conclusion:

"In Phase 2 of our project, we focused on preparing and evaluating machine learning models for sentiment analysis of customer reviews. We meticulously evaluated three powerful algorithms: Random Forest, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN). Our evaluation process involved assessing key performance metrics such as accuracy, precision, recall, and F1-score.

The results of our evaluation revealed that all three models showed promising performance, with Random Forest achieving a perfect performance across all metrics, and SVM and KNN demonstrating strong accuracy and overall effectiveness in predicting customer sentiments.

Moving forward, our objective is to leverage the insights gained from Phase 2 to optimize and fine-tune the selected model further. Additionally, we plan to integrate the model into a real-world application, enabling seamless and efficient sentiment analysis of customer reviews in real-time.

Overall, Phase 2 has laid a solid foundation for the successful implementation and deployment of our predictive model, bringing us closer to achieving our project goals of enhancing customer experience and decision-making through sentiment analysis of customer feedback."